



# Gemini 4 Argon

## Model evaluation

Approach, methodology & results

# Approach

We evaluated Gemini 4 Argon across a range of benchmarks, including agentic coding, ML engineering, computer use, knowledge work, science and math, multimodal capabilities, long-context and cybersecurity.

## Methodology

All Gemini 4 Argon scores are pass @1 except where otherwise noted. "Single attempt" settings allow no majority voting or parallel test-time compute. All of the results are run with the Gemini API with the highest thinking settings unless indicated otherwise below. To reduce variance, we average over multiple trials for smaller benchmarks.

All the results for non-Gemini models are sourced from providers' self reported numbers unless otherwise mentioned below. For GPT-6 Astra, Fable 5.1 and Opus 5.5, we default to reporting maximum thinking/reasoning settings available, but when reported results are not available we use best available reasoning results.

## Additional Details

Our benchmarks span several capabilities as of September, 2026:

- **Knowledge work:**
  - *Vals Index* results are sourced from Vals AI.
  - *AutomationBench* results use the private set and are sourced from the official Zapier [public leaderboard](#).
  - *Vals Finance Agent v2* results are sourced from Vals AI.
  - *Harvey's Legal Agent Benchmark* results are sourced from Vals AI.
- **Coding**
  - *DeepSWE v1.1* results for Gemini 4 Argon are self computed, using a mini-swe agent harness. GPT-6 Astra results are reported from the official [public leaderboard](#), Fable 5.1 and Opus 5.5 results are taken from their respective system cards. We use the highest scoring thinking level for each model as reported by Datacurve on the main leaderboard.
  - *FrontierSWE* results are reported from Proximal's official [public leaderboard](#).
  - *Vibe Code Bench* results are reported from the official Vals AI [public leaderboard](#).
  - *Terminal-Bench 4.0* results for Gemini 4 Argon are self computed, all other models are reported from the official [public leaderboard](#). We use the highest scoring thinking level for each model as reported by the Terminal Bench authors.

- **ML Engineering:**

- *PostTrainBench v1.1* results for all models are self-computed. We use v1.1 with the OpenCode harness for all models and a 10-hour budget on a single NVIDIA H100 GPU. Performance is measured by a weighted aggregate across four base models and seven benchmarks.

- **Science and math**

- *Terminal-Bench Science 0.1* results for Gemini 4 Argon are self computed, with 6x verifier timeout to address timeout issues with verification. All other results are reported from the official [public leaderboard](#).
- *RiemannBench* results are reported from the official Surge [public leaderboard](#).
- *LABBench2* results for all models are self computed. Models are given access to a Linux terminal with pre-installed bioinfo tools, Python, and R, as well as internet access.

- **Long Context:**

- *Graphwalks* results for all models are self-computed. The “Up to 128k” subset uses the 650 items with context length up to 128k tokens. The “256 to 1M” subset uses the 200 GraphWalks problems with context lengths between 256k and 1M tokens.

- **Computer use:**

- *Agent’s Last Exam* results for Gemini 4 Argon are self computed using the default ALE-Claw harness and a 5-hour window, with safety filters enabled. Responses flagged by safety filters are returned as empty strings, though the episode can continue. GPT-6 Astra and Opus 5.5 results are taken from the official [public leaderboard](#). Fable 5.1 results are missing from the public leaderboard so are not included. We report the binary pass rate.
- *OSWorld 2.0* score for Gemini 4 Argon is self computed, maxed over 3 runs with a single attempt per run. Responses flagged by safety filters are returned as empty strings, though the episode can continue. We report the partial score from the offline subset, which has more robust verifiers. We use the github OSWorld 2.0 docker, official evaluator, default 1080p resolution, and max step length of 500. We use the Gemini CUA harness in the OSWorld 2.0 repo, with parallel batch tool calling enabled. We also include a compaction procedure following the same setup as reported in the Opus 5.5 [system card](#). We use pyautogui for actuation with screenshot-only observation space, and for tools we use UI-specific function declarations. Runs were completed with the official OSWorld 2.0 team’s 08.08 patch. GPT Astra results were taken from the [official blog post](#). Anthropic only reports aggregated results for the online and offline subsets combined, so results from Anthropic are not included.

- **Multimodal:**

- *Chartography* results are without tools and taken from the official Surge [public leaderboard](#).
- *LVBench* results are self computed without tools. We use 1FPS for Gemini, 800 frames for GPT-6 Astra and 300 frames for fable 5.1 and 600 for opus 5.5 due to API limitations.
- **Cybersecurity:**
  - *CWE-bench v1* results are taken from the official [public leaderboard](#). Scores are ranked based on pass@1 and ties broken based on pass@4.
  - *Real world vulnerability* is measured using an internal dataset that tests recall over recent, confirmed historical vulnerabilities across a diverse set of programming languages in popular open source projects. Our models ran in an internal version of the Antigravity harness that is not cyber specialized and are given access to the source code.
  - *Wiz Penetration Testing Benchmark* is an internal cybersecurity benchmark measuring ability to write exploits against real web application vulnerabilities, without access to the codebase.
  - *Gray Swan IPI Benchmark* results were sourced from [Gray Swan](#).

# Results

Gemini 4 Argon results as of October, 2026 are below:

|                          | Benchmark                      |                              | Gemini 4 Argon | GPT-6 Astra | Claude Fable 5.1 | Claude Opus 5.5 |
|--------------------------|--------------------------------|------------------------------|----------------|-------------|------------------|-----------------|
| Knowledge work           | Vals Index                     |                              | 68.9%          | 63.1%       | 65.8%            | 67.0%           |
|                          | AutomationBench                | Score                        | 51.3%          | 41.4%       | 31.4%            | 42.5%           |
|                          | Vals Finance Agent v2          |                              | 65.4%          | 53.5%       | 58.9%            | 58.6%           |
|                          | Harvey's Legal Agent Benchmark |                              | 19.6%          | 5.4%        | 6.7%             | 3.8%            |
| Agentic coding           | DeepSWE v1.1                   |                              | 77.9%          | 74.1%       | 67.4%            | 74.2%           |
|                          | FrontierSWE v2                 |                              | 55.0%          | 65.5%       | 56.3%            | 62.3%           |
|                          | Vibe Code Bench                |                              | 91.9%          | 89.6%       | 90.3%            | 90.3%           |
|                          | Terminal-bench 4.0             |                              | 57.4%          | 58.2%       | 57.9%            | 66.4%           |
| ML engineering           | PostTrainBench                 |                              | 45.3%          | 44.3%       | 40.2%            | 49.3%           |
| Science and math         | Terminal-Bench Science 0.1     |                              | 57.6%          | 68.1%       | 52.6%            | 63.3%           |
|                          | LABBench 2                     |                              | 88.8%          | 85.4%       | 68.6%            | 73.1%           |
|                          | RiemannBench                   |                              | 76.0%          | 72.0%       | 65.6%            | 69.6%           |
| Long context             | GraphWalks                     | Up to 128k, BFS (F1)         | 99.7%          | 98.7%       | 91.4%            | 90.6%           |
|                          |                                | 256k to 1M, BFS (F1)         | 84.2%          | 71.8%       | 65.0%            | 66.8%           |
| Computer use             | Agent's Last Exam              | Pass rate                    | 39.5%          | 34.2%       | —                | 38.2%           |
|                          | OSWorld-2.0                    | Offline subset Partial score | 69.2%          | 72.6%       | —                | —               |
| Multimodal understanding | Chartography                   |                              | 71.6%          | 71.0%       | 46.2%            | 66.3%           |
|                          | LVBench                        |                              | 91.7%          | 87.5%       | 79.7%            | 83.7%           |
| Cybersecurity            | CWE-bench v1                   |                              | 68.0%          | 68.0%       | 58.0%            | 67.0%           |

Methodology: [deepmind.google/models/evals-methodology/gemini-4-argon](https://deepmind.google/models/evals-methodology/gemini-4-argon)